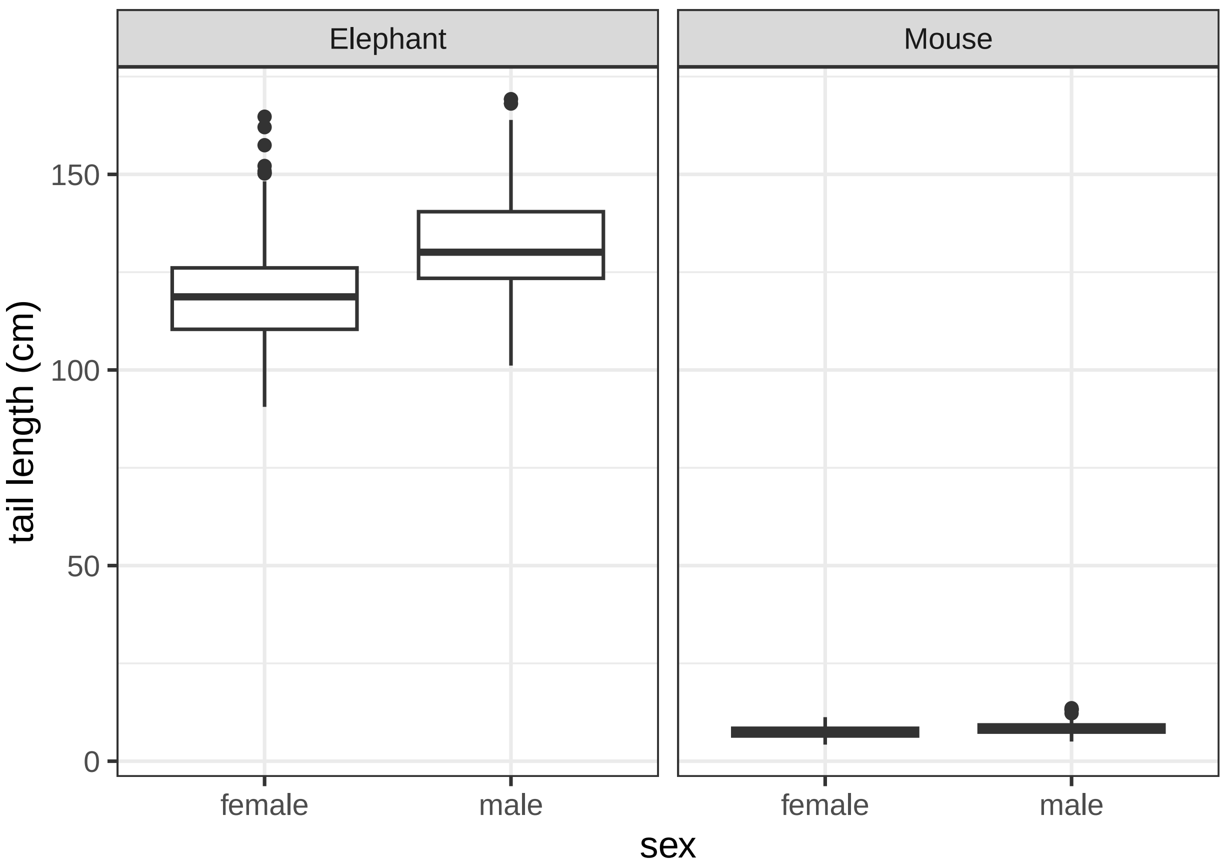
Goals

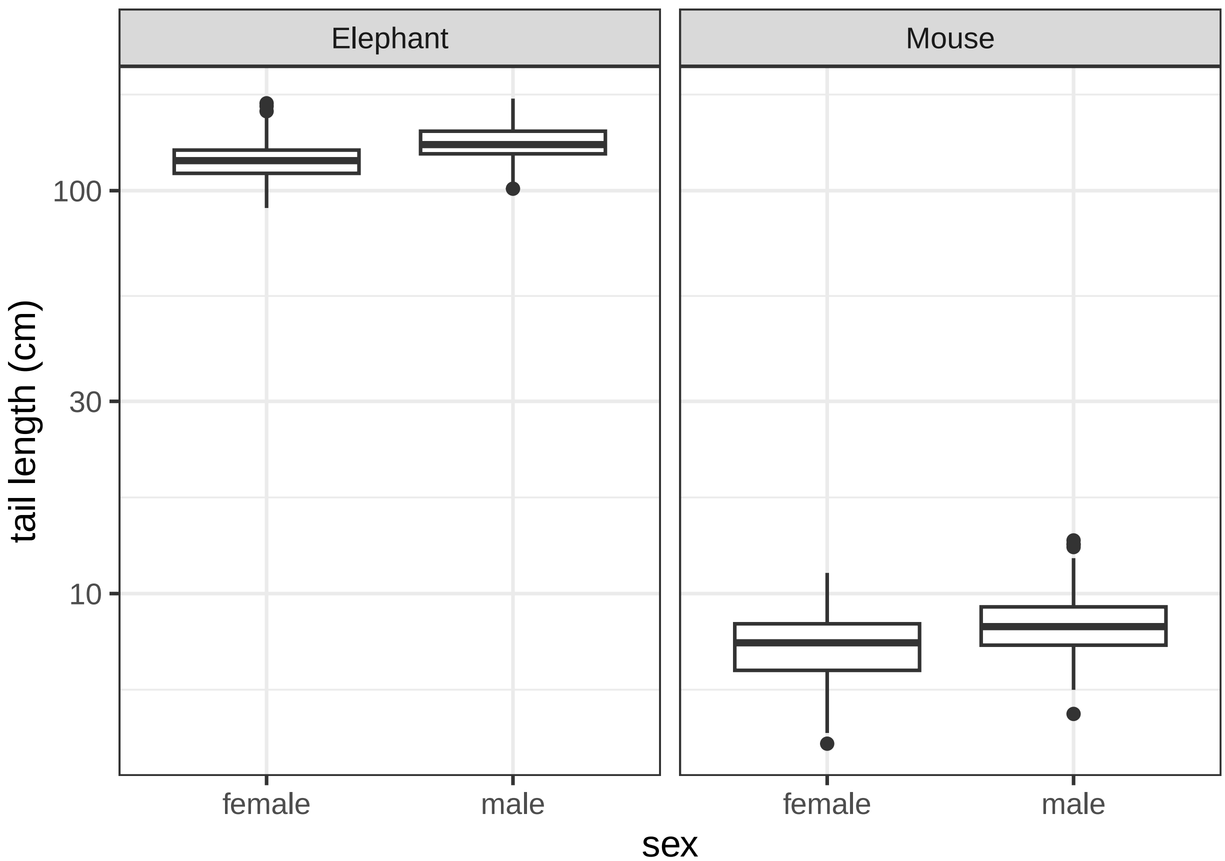
- Introduce some ways of measuring effects
- Distinguish between measures of effect and measures of clarity
- Discuss advantages and disadvantages of standardized (unitless) measures of effect

1 Effects vs clarity

Tail-length data



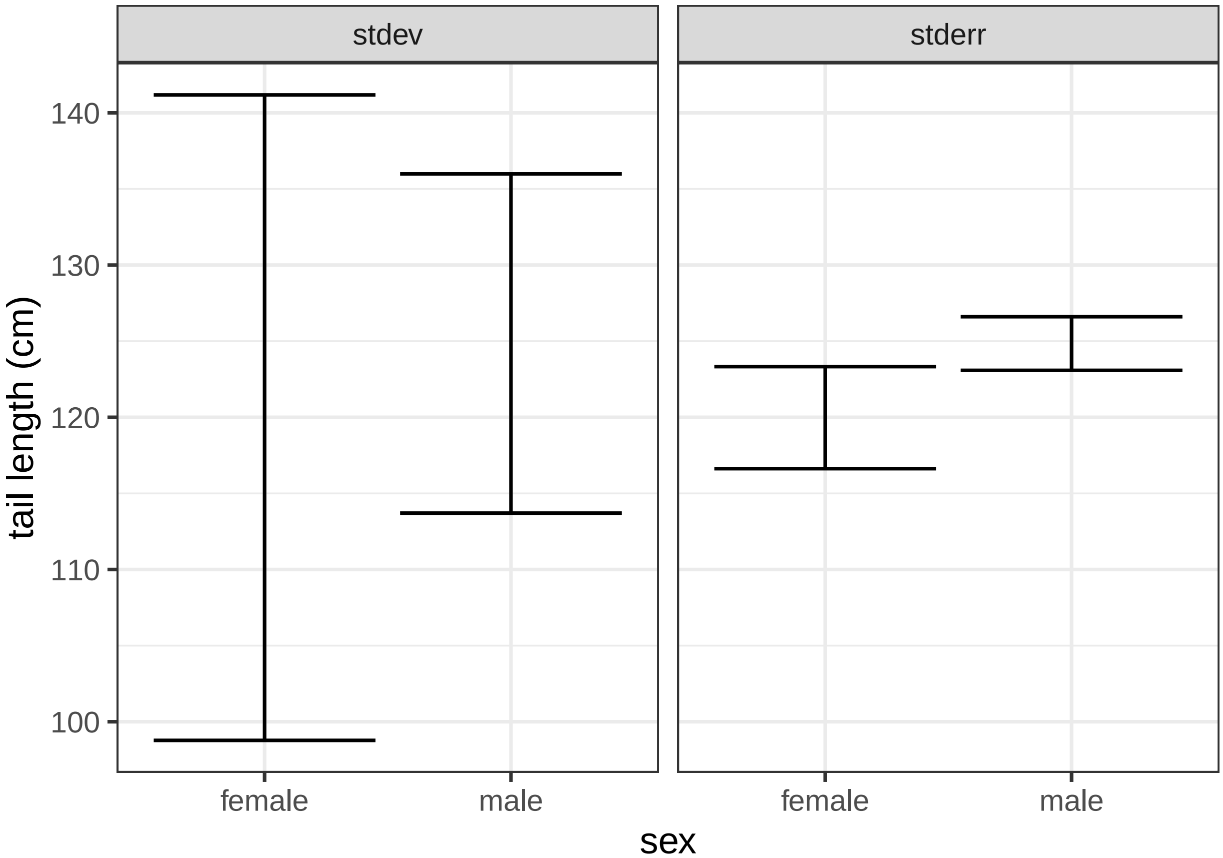
Tail-length data



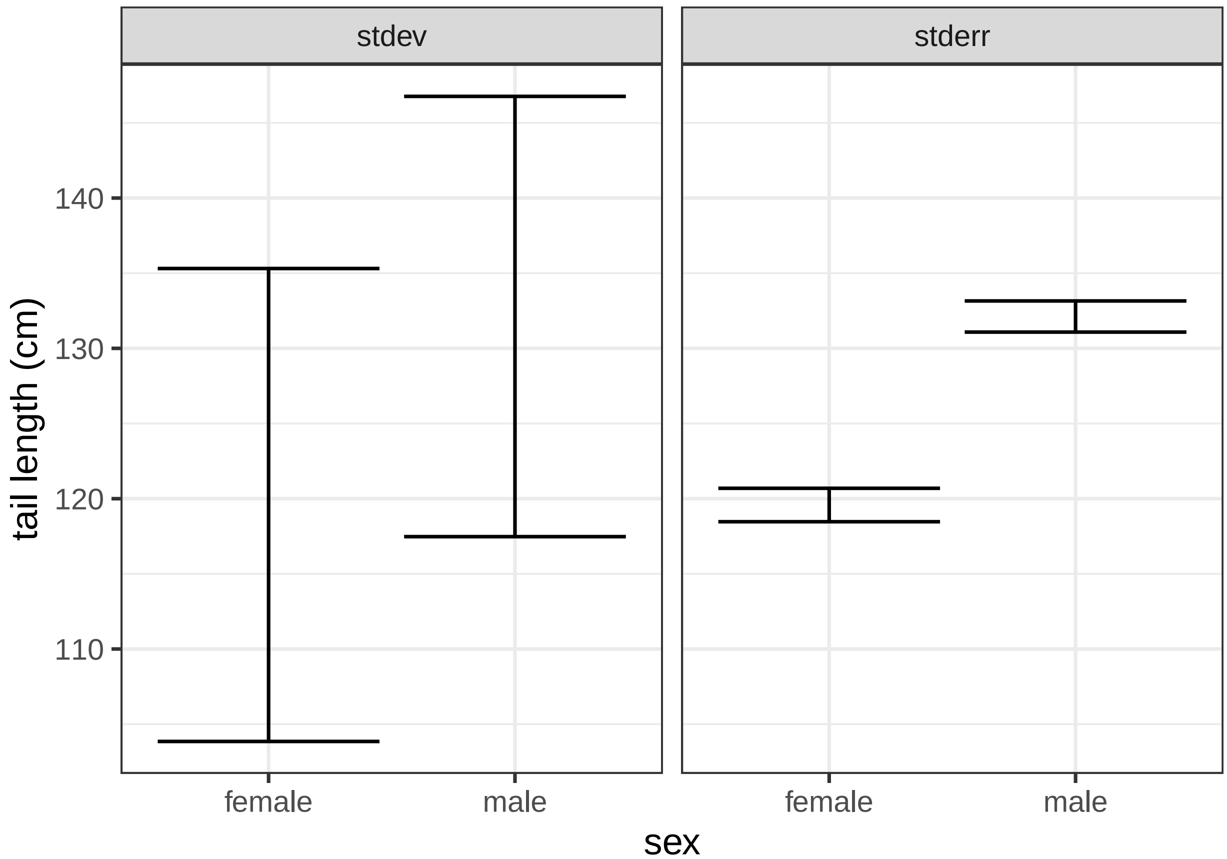
Model summary

```
Call:
lm(formula = tailLength ~ species * sex, data = tails)

Residuals:
    Min      1Q  Median      3Q     Max 
-30.997  -2.401  -0.246   2.156  45.161 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)           119.580      1.080 110.685  < 2e-16 ***
speciesMouse         -112.075      1.528 -73.355  < 2e-16 ***
sexmale                12.538      1.528   8.206 3.24e-15 ***
speciesMouse:sexmale  -11.536      2.161  -5.339 1.58e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Small sample



Large sample



A story

- A very cool biology seminar speaker told us this
  - Vitellogenin is a female hormone, male fish shouldn't have any
  - But we found that males in the disturbed site had *significantly more* vitellogenin
  - The picture showed about 4× increase, but speaker did not emphasize this
- **Answer:** Significance is a measure of clarity, not effect
- **Answer:** 4× is a measure of effect, but is it one that we understand?

2 Absolute effects

Summarizing effects

- In most cases, statistical analysis is focused on a one-parameter effect
  - Difference (or interaction)
  - Slope
- Other cases are more complicated
  - Species distribution
  - Gene expression
  - Life-history tradeoffs

Simple effects

- Difference
  - Mice in the treatment group gained on average 25mg (-2mg, 41mg) more weight than mice in the control group
  - Male fish in the disturbed area had 7.2 ng/ml more vitellogenin than those in the undisturbed area
  - Interaction: Scores in intervention group improved by 2.3 points (1.1pts, 5.0 pts) more than they improved in the control group
- Slope
  - We estimated an increase of 4000 annual deaths per fluA percent positivity (2700d/fluAppos, 5500d/fluAppos)
  - Soybean yield is estimated to increase 50kg/ha/°C in this temperature range (I'm tired of the CIs now, you get the idea)

Units ...

- are critical to interpreting these effect sizes
- Advantages
  - You need to engage with and understand these units
    * This drives biological understanding
    * And can force thoughtful choices about biological importance
- Disadvantages
  - You need to engage with and understand these units
    * Can be difficult, not always relevant

  - You can't compare across different systems
    * Are mice more dimorphic than elephants?
    * Which natural characteristics are under strong directional selection?

3 Standardized effects

Making unitless variables

- Divide by something with the same units
  - Mean, standard deviation, difference
- Take logs
  - **Answer:** You're still dividing by something with the same units
  - **Answer:** Once you log it, it doesn't matter what you chose
- What *kind* of scale do you need for these?

Example: correlation

- What is a correlation (ρ)?
  - **Answer:** It is the slope of the relationship between two variables
  - **Answer:** After both are divided by their sd
- What is R^2?
  - **Answer:** it is literally ρ^2 – terrible nomenclature
- R^2 reflects variation explained and ties to deep stuff about linear modeling
  - e.g., you can add R^2 values from independent predictors

Example: Cohen and Hedges

- ρ is a unitless slope based on standard deviations
- Cohen's d and Hedges g are unitless differences based on standard deviations
  - They calculate pooled sd differently
  - You should prefer Hedges

Proportional change

- If we divide a physical quantity (like tail length) by a group mean, we're now talking about proportional change
- Logs are a somewhat different (and generally more consistent) way to think about proportional change

Proportional change

- Proportional change is a bit weird:
  - If the females are 40% lighter than the males:
  - **Answer:** the males are 66.7% heavier than the females
- Using (natural) logged ratios is a very appealing substitute
  - log(1.67)=0.511; log(0.6)=-0.511
  - JD: log ratios are generally more sensible than proportional change
- Both methods can be used for either differences or slopes

Mix and match

- Phenotypic evolution studies the effect of traits on measures of fitness
- The field decided to construct standardized slopes by
  - Dividing trait measures by their sd
  - Dividing fitness measures by their mean
- Why not?

Other options

- You may have a biological reason for scaling one difference by another difference with the same units
  - e.g., how big is the difference between males and females compared to the difference between different species in a genus?
  - Vitellogenin: how much is the increase in male levels compared to the average female level?

Comparing across contexts

- Unitless effect sizes can be used to compare across contexts
  - Are mice more dimorphic than elephants?
  - Is natural selection for size stronger in trees than giraffes?
- Needs to be done with care
  - Is your difference driven by changes in the denominator?
    - e.g., g tends to be larger in lab populations because variation is less
  - What is a good cutoff of large/medium/small for your particular context and question?

4 Scaling *predictors*

- Sort of split the difference between absolute and standardized
- If we have many predictors for one outcome, it may make sense to keep the units on the outcome only
- If we had one predictor and many outcomes, it might make sense to do the opposite
  - e.g., we might measure the effect on a number of plant traits in units of [1/ppmCO2]
  - Could standardize by mean or sd

Z scores

- Subtracting the mean and dividing by the standard deviation is the "standard" way of standardizing
  - You can also skip the subtraction step
- This allows comparison of the (estimated) effects of different predictors across the study
  - Related to their importance in predicting changes in outcome

Mean standardization

- If you instead divide by the mean you are looking at effects on the scale of the average amount of the predictor
  - Comparing to the counter-factual than an effect is absent

Influenza study

- In the influenza study, we mean-standardized our predictors but kept our outcome as deaths/year
  - **Answer:** Effect size was deaths per year
  - **Answer:** In other words, compared to the counter-factual of no influenza
- For weather variables, it might have made sense to use z-scores instead
  - **Answer:** How much of the variation we *saw* in Chicago was explained by changes in winter temperatures
  - **Answer:** As opposed to how much less death would we expect if there were no cold days in Chicago winters

5 Conclusions

- Differentiate measures of effect from measures of clarity
  - Prefer the former when discussing scientific implications of your result
- Think carefully about whether to standardize
  - Pay attention to units if you keep them
- Think carefully about how to standardize
  - Make sure your interpretations are consistent with your choices

A modest proposal

- One P value is sometimes useful as a summary of a substantive part of an analysis
  - We should never be looking at a lot of P values
- What could we do instead?
  - **Answer:** Distribution of effect sizes
  - **Answer:** Distribution of minimal effect sizes (based on confidence intervals)
- Clarity is often used as a simplistic (and lazy) proxy for something about effect size
- What if I can't avoid this in my field?
  - **Answer:** Do the best you can to move in a good direction
  - **Answer:** Or to reason around it